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By Gardner, of the Derola Medical College.

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ON PREVENTIVE INOCULATION WITH ATTENUATED MICRO-MANA.

By Gardner, of the Derola Medical College.

Received in the Month of Frostfall, F.E. 262.

SUMMARY. Treatment of the micro-mana of the tide-carbuncle with an order ward yields three sorts of material: those whose virulence returns upon removal from the ward, those which multiply no more, and those which still multiply but have lost much of their virulence. Only the third sort is considered here. Its power to cause disease is examined after removal, transport and passage; and a comparison of mortals and sheep, assigned to their groups beforehand in three cities, is reported for the years 245–256, extending through the ten-year tide of 254. In all three cities the inoculated suffered fewer attacks of the tide-carbuncle. Ill effects following inoculation are recorded separately. The potency of the preparation is determined by its actual protection of beasts, each batch being tried alongside a standard sample; fresh samples are examined upon arrival in another city. Attenuation, protection and potency each require their own evidence.

I. WHAT MUST BE KNOWN BEFORE INOCULATION

Mercer taught us that the tide-carbuncle arises from living micro-mana. [1] When he killed the micro-mana, inoculation no longer produced the original disease; but that a sheep did not fall sick then does not show that it would remain well when it later encountered living micro-mana. Earlier attempts at inoculation approached the matter from the other end: material from the sick was put into those who had not been sick. Some suffered only a light illness and indeed escaped further attacks; others suffered severely, and some died. When material bearing one name yields such different results, it is not enough to advise the physician to be more careful with the quantity.

My attention was first drawn to the flocks outside the city. In several folds regularly covered by small wards, fewer sheep fell sick in the tide years; yet some soon fell sick when moved to other folds. Merely adding the old fold-books to the new might suggest that the ward had left the sheep with protection. Inquiry into each beast often showed, however, that within the ward it had never encountered enough micro-mana to cause disease, and first did so after its removal. Winslow teaches that an agreement found in the registers is a reason to undertake a trial, not evidence that the trial has already been made. [2] This is especially true here.

The questions of this paper are whether the virulence of micro-mana can be diminished and remain so after removal from the ward, and whether such living things, when inoculated, can lessen illness in a later plague. Unless these two things are separately proved, preventive inoculation merely brings a future danger nearer.

II. ATTENUATION AND TEMPORARY SUPPRESSION

I divided isolated micro-mana of the tide-carbuncle into several portions, placed them within order wards, and numbered each. Fewer points of light and slower division within the ward were recorded only as observations during treatment; judgement was reserved until the material had left it. Portions which multiplied no more after removal and left only fine crystals were kept apart as killed material. Those which resumed their multiplication and caused disease as readily as the original stock were rejected. What follows concerns one line which still multiplied but caused less disease. I shall call it line A.

To be alive means to divide, and for the products of that division to divide again; the brightness of a point of light is no proof. Diminished virulence means that, under the same conditions of observation, comparable numbers of living micro-mana given to healthy beasts of the same kind cause fewer and milder illnesses. It does not mean that scarcely any living things remain in the dish. The College kept the identities of the portions; only code marks accompanied them to the trial, and the physician examining the sheep completed his record of their symptoms before the identities were unsealed and compared.

Table 1 gives one comparison, using a separate group of twenty-four sheep for each material and observing them for a full month. Before the trial, none had a history of the tide-carbuncle, and the groups were alike in age and bodily condition. The untreated stock and blank fluid for raising were tried at the same time. To count as the tide-carbuncle, an illness had to include fever and a black carbuncle, with corresponding micro-mana isolated from the diseased part. Redness at the place of inoculation alone was not counted as the original disease.

TABLE 1. Tide-carbuncle following inoculation after removal of the material from the ward

Material inoculated	*Sheep*	*Tide-carbuncle*	*Deaths from it*
Untreated stock	24	22	8
A portion quiet within the ward, restored after removal	24	20	7
Line A, after removal	24	1	0
Line A, transported over a long route, then passed ten times in unwarded dishes	24	1	0
Line A, passed twenty times in unwarded dishes	24	1	0
Killed material	24	0	0
Blank fluid for raising	24	0	0

The transported branch travelled from Derola through Varen to Onais and back aboard ordinary cargo ships; the untransported branch remained at Derola. The material of the tenth and twentieth passages still multiplied. The three cases in the three groups testing line A were all mild, and none had the severity seen with the original stock. This does not prove that virulence can never return. It proves only that line A did not regain its former virulence during the transport and passages it underwent. Each later batch must still be examined in the same way, and cannot be released upon the reputation of the first dish.

I next took three other groups of sheep, twenty-four in each, and first gave them line A, killed material and blank fluid respectively. None developed the tide-carbuncle during the following month. All three groups were then tried with the same disease-producing material. Three sheep fell sick in the line A group, eighteen in the killed-material group, and twenty in the blank group. This comparison establishes that line A had given protection. The killed portion had caused no disease when first given, yet it did not protect as line A did. This judgement concerns only the killed material used on this occasion, and does not decide the merits of every other manner of preparation.

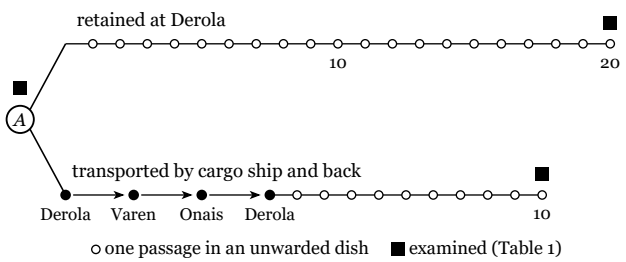


Fig. 1. The keeping and dispatch of line A: the branch retained at Derola, the branch transported through Varen to Onais and back, and the three examinations made after removal from the ward, at the tenth passage after transport and at the twentieth passage at Derola

III. HOW THE REGISTERS OF THE THREE CITIES WERE BEGUN

The later trial upon a small flock cannot take the place of more than ten years of records in the cities. After the tide of 244, I agreed with the physicians of Onais and Varen that we should begin our registers together in the following year and make our general report after the next ten-year tide. We followed Winslow's form of case-register, with separate entries for race, age, old injuries, experience of spellcasting, former illnesses, the material and quantity given, the date of inoculation, and the findings at each examination. [2] The batch number and the record of the examination on arrival were also copied into it, so that, should a batch later prove faulty, every recipient could be traced.

The comparison among mortals was confined to adult residents who had never had the tide-carbuncle and were well at entry. Residents of mixed blood were included in the mortal register, but their race remained separately entered, preserving the original information. Before lots were drawn, all entrants knew the results of

the earlier sheep trials and that illness remained possible; they volunteered for the comparison of inoculation with no inoculation and agreed to be followed. In each city, residents were first arranged in small comparable sets by quarter, occupation, age and bodily condition; sealed lots then assigned the places among the inoculated and uninoculated, one hundred and twenty to each group in each city. The lots were opened before inoculation; a person could not be moved to the other side after he had fallen sick.

Both groups had the same protection of the city's quarantine and wards, and received the same care when ill. Inoculation was completed in 245, without further inoculations during follow-up; the comparison group remained uninoculated when the registers were closed. The original disease among mortals was observed only in its natural outbreaks; no disease-producing material was given to them for a later trial. The three cities used the same rule for judging illness. The visiting physician first recorded the symptoms, and another person then checked the original group assignment. Suspected but unresolved illnesses remained in the column for doubtful cases; they were not erased according to the physician's confidence in inoculation. They are not counted as confirmed cases in the table.

Sheep were likewise registered in all three cities, forty in each group in each city, and matched and assigned by lot beforehand according to origin, age and conditions of keeping. These were the sheep for long observation, not those used in Table 1 or in the later trial. Their inoculation too was performed only in 245, with no repetition or change of groups afterwards. The folds were interspersed, both groups were tended by the same rules, and removals, sales and deaths were followed up. When a mortal moved to either of the other cooperating cities, his register went with him, but he remained in the count of the city where he had entered. Only a removal beyond the reach of inquiry was counted as a loss to follow-up. A blank left by someone lost cannot be filled in as "not sick". The three cities retain their own books of assignments, batch numbers and yearly cases, so that the records can be checked against one another. [3]

IV. THROUGH ONE TEN-YEAR TIDE

Inoculation took place in 245, and the registers were brought together at the end of 256. After assembling the registers, we first settled the methods for determining potency across cities, and submitted the two sets of findings together in 262. Table 2 gives the results among mortals in the three cities. In each of the three periods, only first attacks are entered; each person appears in one period alone. Those who died are already among the cases and are not added a second time. Each row began with one hundred and twenty persons; the number was not replaced by that of those who afterwards remained in the city. The losses are persons who had no known attack of the tide-carbuncle during observation and whose whereabouts were ultimately lost, or whose doubtful illness was never resolved.

Table 2. First attacks of the tide-carbuncle among mortals in three cities; 120 persons originally entered in each row

City and group	*245–253*	*Tide year 254*	*255–256*	*Deaths from tide-carbuncle*	*Lost*
Derola, inoculated	1	3	1	0	3
Derola, uninoculated	9	23	3	5	4
Onais, inoculated	1	4	0	1	5
Onais, uninoculated	8	21	3	4	5
Varen, inoculated	1	2	1	0	4
Varen, uninoculated	9	22	3	5	5

Over the whole period, there were fourteen known cases among the three hundred and sixty inoculated, and one hundred and one among the three hundred and sixty uninoculated; deaths from the tide-carbuncle numbered one and fourteen respectively. A further six in the inoculated group and seven in the uninoculated died from other established causes. The date of death was entered and observation stopped on that date; they were not treated as healthy persons followed to the end. The unresolved doubtful illnesses, four among the inoculated and six among the uninoculated, are already included in the lost column. No person has been entered twice as two sorts of loss.

Before the tide began in 254, three of the inoculated had already been sick, four had been lost without a previous attack, and three had died from other causes without a previous attack. The corresponding numbers among the uninoculated were twenty-six, five and three. These three categories do not overlap. After their removal, three hundred and fifty inoculated and three hundred and twenty-six uninoculated persons remained in the register without a previous attack. All have complete records for the tide year. The nine and sixty-six new cases of that year arose from these two numbers. The difference between the groups was therefore not confined to ordinary years; nor did the tide year sweep away all their earlier protection.

It may be said that all twelve inoculated persons in the lost column had fallen sick without being entered as cases. Even if all twelve are counted as tide-carbuncle cases, the known and supposed cases together number only twenty-six, whereas on the uninoculated side the known cases alone number one hundred and one. This reckoning cannot recover the missing records, but it shows that these omissions cannot reverse the difference observed. To say precisely how many illnesses were prevented in every hundred, however, one cannot pretend that everyone was followed to the last day and settle the matter merely by dividing the totals of two rows.

The sheep are recorded in Table 3, with the same periods and manner of counting. Over the whole period, twenty inoculated and twenty-one uninoculated sheep died from old age, injury or other established causes apart from the tide-carbuncle. Those which could not be traced after removal are separately entered as lost; new sheep were not brought in to replace them.

TABLE 3. First attacks of the tide-carbuncle among sheep in three cities; 40 sheep originally entered in each row

City and group	*245–253*	*Tide year 254*	*255–256*	*Deaths from tide-carbuncle*	*Lost*
Derola, inoculated	1	3	1	1	1
Derola, uninoculated	8	14	2	8	2
Onais, inoculated	1	2	0	1	2
Onais, uninoculated	7	13	3	7	1
Varen, inoculated	2	3	1	1	1
Varen, uninoculated	9	13	2	7	2

Fourteen of the one hundred and twenty inoculated sheep fell sick during the whole period, against seventy-one of the one hundred and twenty uninoculated. Before the tide of 254, four of the inoculated had already been sick, two had been lost without a previous attack, and twelve had died from other causes without a previous attack. The corresponding numbers among the uninoculated were twenty-four, two and twelve, again without overlap between the three categories. Thus one hundred and two and eighty-two respectively remained in the register without a previous attack, and all were observed throughout that year; eight and forty of them fell sick. No city gave a result contrary to that of the other two. Yet the evidence extends only through one ten-year tide. To carry one tide forward into ten is to read a register which has not yet been written.

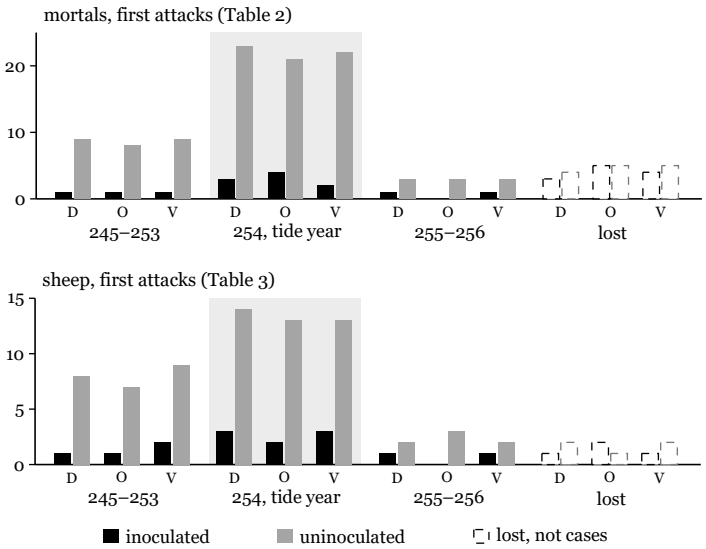


Fig. 2. First attacks among mortals and sheep in the three periods after inoculation, with the ten-year tide of 254 shown separately; mortals and sheep in separate panels, and losses not entered as zero cases. The three periods are of nine, one and two years, and the heights are accumulated cases

V. SOME CASES TO BE READ BESIDE THE TOTALS

CASE I. R, a porter of Derola, was thirty-four at entry and had never had the tide-carbuncle. He was inoculated in 245. His arm swelled on the next day, fever began on the third, and by the fifth the fever had passed; no black carbuncle appeared. In 254 he was still working at the same quay. Two uninoculated persons living with him fell sick, but R did not. To copy only the record after the fifth day would erase an episode of fever; to call that fever the tide-carbuncle would represent protection as requiring the original disease first. Neither account agrees with what was observed.

CASE II. M, a herdsman of Onais, was forty-one at entry. After inoculation in 245 he had only local redness. In 254 he developed fever and a carbuncle; the micromana isolated agreed with those of the tide-carbuncle then present in the city, and he subsequently died. He is the death in the inoculated group of Table 2. His original batch number, the examination record of that year and the other recipients of the same batch have all been checked. No difference sufficient to explain this case was found. That most received protection does not allow us to say that this man suffered from some other illness.

CASE III. An inoculated sheep of Varen was recorded in the fold-book as well after inoculation, but developed the tide-carbuncle in 254. Its keeper first declared that the transported material had lost its effect. Examination of the arrival record for that batch and of the other sheep's records supplied no evidence for this judgement. The sheep recovered. One failure alone cannot establish that the whole batch is ineffective; equally, one sheep's escape cannot establish that the whole batch is effective.

Among the three hundred and sixty recipients, sixty-three had local redness and swelling in the first month after inoculation, and seventeen had fever. Eleven of the seventeen also had redness and swelling, so the two numbers cannot be added to count persons. Two were unable to work for more than seven days because they were unwell; both are among those already counted. Six of the uninoculated group also had fever during the same period. There was no confirmed tide-carbuncle or death among the inoculated in that month; all later illnesses and deaths continued to be entered under the original groups. Among inoculated sheep in the first month, twenty-one had redness and swelling and nine had fever, six of the latter also having redness and swelling; there was no confirmed tide-carbuncle or death.

After the first month, discomforts and newly observed harm continued to be entered at every visit in both groups until the registers were assembled in 256. The original group assignments were preserved, and the observations of each year are recorded in full in the collected registers. [3] I therefore say only that this material greatly diminishes the danger of severe disease from inoculation, not that the danger has ceased. The mild illnesses of Table 1, the discomforts following inoculation and the later cases despite protection must be sent to physicians along with the favourable results. Nor do the existing records allow each of the fourteen later cases among residents to be assigned with certainty to natural infection or to some connection with the earlier inoculation. Retained samples and the records of many years must still be kept.

VI. HOW THE POTENCY OF A BATCH IS DETERMINED

Long transport presents another circumstance liable to mislead. A portion reaching Onais still showed living things and retained its former elemental affinity, yet its protection was insufficient. Had only the points of light and the constituents been examined, it would have been issued as before. I therefore separate the examination of disease-producing power from that of protective potency. The former must first be passed; no excellence in the latter can make good a failure in it.

For the material of this paper, the endpoint of potency is fixed as follows: in the prescribed sheep trial, after preventive inoculation, at least three-quarters must remain free of the tide-carbuncle when later tried with disease-producing material. The duration of observation, the rule for judging disease, the origin and bodily condition of the sheep, and the material used in the later trial must all be fixed beforehand. Each trial includes an uninoculated group, which must show that the later test can indeed cause disease. If most of them also remain well, the trial cannot determine potency; the preparation under examination cannot on that account be called excellent.

Every batch is tried at the same time as the standard sample. A batch of this year is not compared with a number in last year's book. Several quantities of each are tried to find the amount required to reach the same endpoint. If the amount required of the standard is s, and that of the batch is b, the relative potency of the batch is e:

$$e = \frac{s}{b} \tag{1}$$

If twice as much of the batch is needed to obtain the standard's protection, its potency is half that of the standard. Here a "portion" is an agreed equal volume, used only to compare materials. It must not be redefined as a number of living things, for that would remove from the reckoning the very difference to be examined. Each determination is repeated with other sheep, and Table 4 gives only the smallest quantity tried which reached the endpoint on both occasions. Results at neighbouring quantities must also agree. If increasing quantities do not produce a similar progression of protection, no relative number is forced upon them; the batch is held and examined again.

Table 4. Relative potency at the same endpoint; quantities in agreed portions

Batch and place	*Standard required, s*	*Batch required, b*	*Relative potency, e*	*Decision*
262 A, Derola, before dispatch	1.00	1.00	1.00	Accepted
262 A, Onais, after arrival	1.00	1.25	0.80	Accepted
262 B, Derola, before dispatch	1.00	1.25	0.80	Accepted
262 B, Onais, after arrival	1.00	2.00	0.50	Not used
262 C, Derola, before dispatch	1.00	2.00	0.50	Not dispatched

Before dividing the batches, we agreed not to use any with less than four-fifths of the standard's potency. This limit was fixed within the range of quantities and protection already tried, and applies only to this preparation. Batch B fell below it after arrival. Its departure mark could not authorize its use, nor could twice the quantity be improvised, passing the loss during transport on to the recipient. The batch number, seals, stops on the journey and arrival date are recorded together. Until a satisfactory arrival examination is entered, the whole batch is held.

VII. WHO EXAMINES THE STANDARD

The standard too may deteriorate. If it and the batch grow weaker together, a relative number still close to one does not show that either remains as it was. The Onais Hospital therefore keeps separate samples for verification and independently repeats the prescribed protective trial each year. It records both the comparison with the Derola standard and the actual amount required to reach the protective endpoint. If both require more than the old records show, the matter must be stopped and examined even though the two remain equal. An irregularity in storage is investigated without waiting until the year's end.

When a standard is replaced, matched groups must be drawn from the same set of sheep, so that old and new standards are compared side by side with the same later test, and the comparison repeated. Writing “standard” upon a new bottle does not entitle it to replace the old. If one and a quarter portions of the new are needed to give the protection of one portion of the old, each new portion is worth only four-fifths of an old one. Its value must be assigned accordingly, or other material chosen; this fifth cannot be erased from the records of successive years. Relative numbers subsequently obtained with the new standard must be multiplied by these four-fifths before entering the potency account kept in the old standard’s units. The transfer must be completed before the old standard is exhausted. Without an overlap for comparison, the former measure of potency has been broken, and one cannot still claim to use it.

Onais and Varen accept deliveries by their own examinations under these rules and send back the records. The sheep, the physicians judging disease and the record-books belong to the receiving city. Only when the cities can reproduce comparable potency and long-term results pointing in the same direction can the material be used across them. Any future preparation claiming protection should first state what it is to protect against, and then establish its own standard. Measurements of elemental affinity and constituents may detect errors; they cannot replace this test of effect.

VIII. CONCLUSION

Through the removal, transport and passages examined, line A remained alive without regaining its former virulence. When it was used for preventive inoculation, mortals and sheep in all three cities suffered less tide-carbuncle, and protection remained evident through the next ten-year tide. We may thus join the ward outside the body to the protection of the recipient himself, without expecting a ward stretched over the whole city to keep every danger from every person.

After inoculation, sick travellers from plague districts must still be isolated where they arrive, and ships and beasts remain subject to quarantine. The inoculated must also remain in the registers. We do not yet know in what year protection will diminish; this assembled register cannot authorize the omission of those to come.

I ask the physicians of every city, when they examine this method, to send the records of ineffective batches, losses to inquiry and unfavourable outcomes after inoculation along with the rest. The grounds on which an inoculating material is allowed into use cannot be decided by its best performance alone.

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